Distribution Theory and Sobolev Spaces

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Motivation

Burger's Equation

Consider the following equation:

$$\begin{cases} u_t + uu_x = 0, x \in \mathbb{R}, t > 0, \\ u(x,0) = u_0(x). \end{cases}$$

.

Motivation (Continued)

Initial Condition

We consider the initial function u_0 as follows:

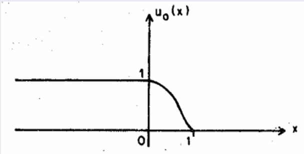

Figure: Initial Function

Motivation (Continued)

Characteristic Curves

The Corresponding Characteristic Curves are :

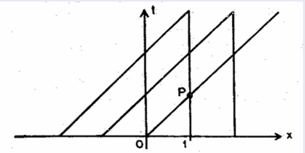

Figure: Characteristic Curves

Motivation (Continued)

Study of PDEs

When we discuss a PDE, its solution should be differentiable as many times as the order of the equation (in classical sense) and it must satisfy the equation everywhere in space.

Problems in Classical sense

Like the previous example, several interesting equations which model physical phenomena fail to possess such property and thus prevent us to study them classically.

Goal

Generalization of the notion of solution

Similarly, many problems can not be solved classically, motivating the need of generalizing the notion of solution of PDEs.

Differentiation in weak sense

The need of generalization of the solution of PDEs motivates the need of generalizing the notion of differentiable functions, leading to study of Distributions.

Discussion in Suitable Spaces

For the study of these generalized notions, we need a suitable spaces, as we will introduce, namely, Sobolev spaces.

Space of Test Functions

Test Functions

Let Ω be an open set in $\mathbb{R}^n$. Then, $\mathcal{D}(\Omega)$, space of C^∞ functions with compact support is called the space of test functions.

Space of Test Functions (Continued)

Example

Consider the function,

$$\phi(x) = \begin{cases} \exp\left(\dfrac{-36}{36 - x^2}\right), & |x| < 6, \\ 0, & |x| \geq 6. \end{cases}$$

Space of Test Functions (Continued)

Figure

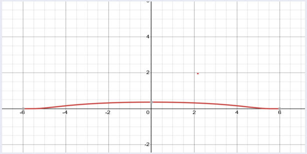

Figure: Graph of the function

This is a C^{∞} function with support $[-6, 6]$. Hence, it is a test function.

Space of Test Functions (Continued)

Convergence in $\mathcal{D}(\Omega)$

A sequence of functions ϕ_m in $\mathcal{D}(\Omega)$ is said to converge to 0 if there exists a fixed compact set $K \subset \Omega$ such that $\text{supp}(\phi_m) \subset K$ for all m and ϕ_m and all its derivatives converge uniformly to zero on K.

Definition

A linear functional T on $\mathcal{D}(\Omega)$ is said to be a distribution on Ω if whenever $\phi_m \to 0$ in $\mathcal{D}(\Omega)$, we have $T(\phi_m) \to 0$.

Space of Distributions

The dual space of the space of test functions, i.e. $\mathcal{D}'(\Omega)$ is called the space of distributions.

Example

Given any $f \in L^1_{loc}(\Omega)$, define $T_f : \mathcal{D}(\Omega) \to \mathbb{R}$ by,

$$T_f(\phi) = \int_\Omega f\phi,$$

where, $L^1_{loc}(\Omega) = \{f : \Omega \to \mathbb{R} \,\big|\, \int_K |f| < \infty$ for any $K \subset \Omega$, K being compact$\}$.

Differentiation of Distributions

Distribution Generated by C^1 function

Let $T \in \mathcal{D}'(\mathbb{R})$. If $T = T_f$, f being a C^1 function, then for $\phi \in \mathcal{D}(\mathbb{R})$,

$$T_{f'}(\phi) = \int_{\mathbb{R}} f'\phi = -\int_{\mathbb{R}} f\phi' = -T(\phi').$$

In this case, we generalize this as,

$$T'(\phi) = -T(\phi'), \phi \in \mathcal{D}(\mathbb{R}).$$

Iterating this, we get,

$$T^{(k)}(\phi) = (-1)^k T(\phi^{(k)}).$$

Differentiation of Distributions (Continued)

Heaviside Function

Consider the function,

$$H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Then, $T'(\phi) = -T(\phi') = -\int_0^\infty \phi'(x)dx = \phi(0) = \delta(\phi)$.

Observation

The distribution derivative is not L^p function, even it is not a function itself, leading to discuss a space where this property is satisfied.

Sobolev Spaces

Definition

Let $m > 0$ be an integer and let $1 \leq p \leq \infty$. The Sobolev Space $W^{m,p}(\Omega)$ is defined by $W^{m,p}(\Omega) = \{u \in L^p(\Omega) | D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \leq m\}$.

When $p = 2$, we denote it by,

$$H^m(\Omega) = W^{m,2}(\Omega).$$

Sobolev Spaces (Continued)

Theorem

$(H^m(\Omega), ||f||_m)$ is a Banach space where,

$$||f||_m = (||f||^2_{L^2(\Omega)} + ||Df||^2_{L^2(\Omega)})^{\frac{1}{2}}.$$

Proof

Let $\{f_k\}$ be a Cauchy sequence in $H^m(\Omega)$. So, for given $\epsilon > 0$ be arbitrary, $\exists\ n_0 \in \mathbb{N}$ such that

$$||f_k - f_l|| < \epsilon \ \forall\ k, l \geq n_0.$$

Sobolev Spaces (Continued)

Proof (Continued)

Therefore,

$$||f_k - f_l||_{L^2(\Omega)} < \epsilon \;\&\; ||Df_k - Df_l||_{L^2(\Omega)} < \epsilon.$$

Since, $L^2(\Omega)$ is complete, $f_k \to f$, $Df_k \to g$ in $L^2(\Omega)$. To prove, $g = Df$.
Now,

$$\int_\Omega f_k \phi' = -\int_\Omega Df_k \phi$$

$$\Rightarrow \lim_{k\to\infty} \int_\Omega f_k \phi' = -\lim_{k\to\infty} \int_\Omega Df_k \phi$$

$$\Rightarrow \int_\Omega f \phi' = -\int_\Omega g \phi$$

using Cauchy-Schwartz inequality and the. So, the space is complete with the given norm. Hence, it is a Banach Space.

Sesquilinear Forms

Definition

Let X and Y be vector spaces, over the same field K $(= \mathbb{R}$ or $\mathbb{C})$. Then a sesquilinear form h on $X \times Y$ is a mapping,

$$h : X \times Y \to K$$

such that for all $x, x_1, x_2 \in X$ and $y, y_1, y_2 \in Y$ and all scalars $\alpha, \beta \in K$,

$$\begin{aligned} h(x_1 + x_2, y) &= h(x_1, y) + h(x_2, y), \\ h(x, y_1 + y_2) &= h(x, y_1) + h(x, y_2), \\ h(\alpha x, y) &= \alpha h(x, y), \\ h(x, \beta y) &= \overline{\beta} h(x, y). \end{aligned}$$

Sesquilinear Form (Continued)

Norm

If h is bounded, the number

$$||h|| = \sup_{x,y \neq 0} \frac{|h(x,y)|}{||x||\ ||y||}$$

is called the Norm of h.

Example

Every complex inner product is a sesquilinear form.

Observation

If $K = \mathbb{R}$, then h is called bilinear since it is linear in both arguments.

Riesz Representation Theorem

Statement

Let H_1, H_2 be Hilbert spaces and $h : H_1 \times H_2 \to K$ a bounded sesquilinear form. Then h has a representation

$$h(x, y) = \langle Sx, y \rangle$$

where $S : H_1 \to H_2$ is a bounded linear operator. S is uniquely determined by h and has norm

$$||S|| = ||h||.$$

Riesz Representation Theorem (Continued)

Sketch of Proof

- Consider $\overline{h(x,y)}$. This is linear in y.
- for fixed x, use Riesz Representation theorem for y to show $h(x,y) = \langle z, y\rangle$
- Define an operator $S : H_1 \to H_2$ by $z = Sx$. So, $h(x,y) = \langle Sx, y\rangle$.
- S is linear.
- $||S|| = ||h||$.
- Consider another such operator T and prove $S \equiv T$.

Lax-Milgram Theorem

V-elliptic bilinear form

Let $a : V \times V \to \mathbb{R}$ be a bilinear form. It is V-elliptic if $\exists\ \alpha > 0$ such that

$$a(v, v) \geq \alpha ||v||^2\ \forall\ v \in V.$$

Lax-Milgram

Let V be a Hilbert space and $a(.,.)$ be a continuous V elliptic bilinear form. Then given $f \in V$, $\exists!\ u \in V$ such that

$$a(u, v) = \langle f, v \rangle\ \forall\ v \in V.$$

Dirichlet Problem for Elliptic Operators

Second Order Elliptic Operators

Le $\Omega \subset \mathbb{R}^n$ be a bounded open set. Consider the problem,

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega, \\ u &= 0 \text{ on } \partial\Omega. \end{aligned} \tag{1}$$

Let $u \in H_0^1(\Omega)$ be such that it satisfies

$$\int_\Omega \nabla u . \nabla v = \int_\Omega fv, \text{ for every v} \in H_0^1(\Omega). \tag{2}$$

Then, u is a weak solution of (1).

Dirichlet Problem for Elliptic Operators (Continued)

Theorem

Let Ω be a bounded open set and $f \in L^2(\Omega)$. Then there exists a unique weak solution $u \in H_0^1(\Omega)$ satisfying (1).

Sketch of Proof

- Set $V = H_0^1(\Omega)$ and $a(u, v) = \int_\Omega \nabla u . \nabla v$.
- Show that $a(.,.)$ is continuous.
- $a(.,.)$ is V-elliptic by Poincaré's inequality and the result follows by Lax-Milgram.

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- Kreyszig, E. (1991). Introductory functional analysis with applications. John Wiley & Sons.

Thank You!